ass10_7331

March 24, 2026

[1]:
```python
import numpy as np
from sklearn.datasets import load_iris
import pandas as pd

iris = load_iris()

df = pd.DataFrame(data=iris.data, columns=iris.feature_names)
```

[2]:
```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 4 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   sepal length (cm)  150 non-null    float64
 1   sepal width (cm)   150 non-null    float64
 2   petal length (cm)  150 non-null    float64
 3   petal width (cm)   150 non-null    float64
dtypes: float64(4)
memory usage: 4.8 KB
```

[3]:
```python
df.head()
```

[3]:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2

[4]:
```python
df.describe()
```

[4]:

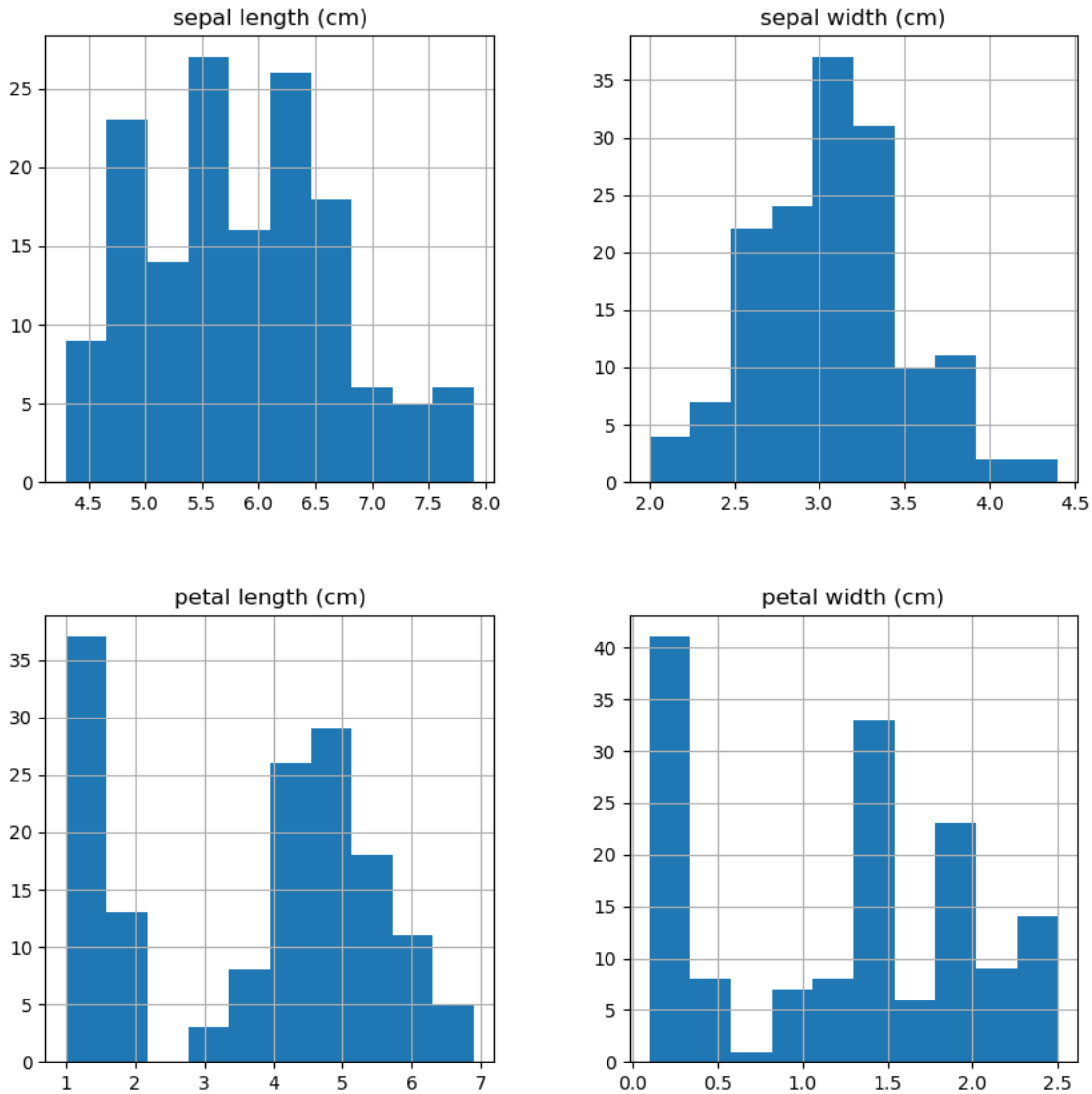
	sepal length (cm)	sepal width (cm)	petal length (cm)	\
count	150.000000	150.000000	150.000000	
mean	5.843333	3.057333	3.758000	
std	0.828066	0.435866	1.765298	
min	4.300000	2.000000	1.000000	

```
25%               5.100000          2.800000           1.600000
50%               5.800000          3.000000           4.350000
75%               6.400000          3.300000           5.100000
max               7.900000          4.400000           6.900000

       petal width (cm)
count        150.000000
mean           1.199333
std            0.762238
min            0.100000
25%            0.300000
50%            1.300000
75%            1.800000
max            2.500000
```

[5]:
```
import matplotlib.pyplot as plt
```

[6]:
```
df.hist(figsize=(10, 10))
plt.show()
```



[7]:

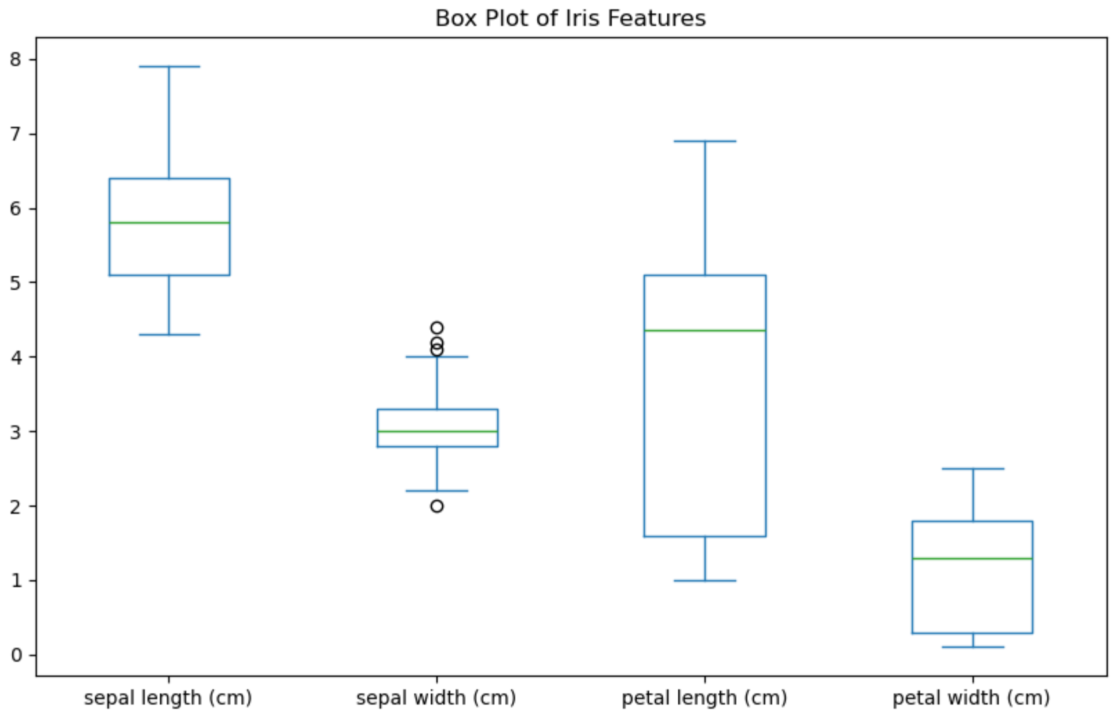
```
df.isnull().sum()
```

[7]:

```
sepal length (cm)    0
sepal width (cm)     0
petal length (cm)    0
petal width (cm)     0
dtype: int64
```

[43]:

```
df.plot(kind='box', figsize=(10, 6))
plt.title("Box Plot of Iris Features")
plt.show()
```


Box Plot of Iris Features
8
7
6
5
4
3
2
1
0
sepal length (cm)
sepal width (cm)
petal length (cm)
petal width (cm)

```
[ ]:
```